

Focal Length Is Everything

The paper establishes two critical observations:

O1: Sensor/pixel size does not matter.

Only the ratio of focal length to pixel size matters, and this is captured by the focal length in pixels f_{px} .

O2: Focal length is vital for metric depth.

Different focal lengths cause different depth scales. The network must compensate, or depth predictions collapse across diverse cameras.

Normals are metric-agnostic:

Surface normals depend only on local geometry, not on absolute depth scale. This

